

Multiple Linear Regression

Theory Questions

TTK4260

Question 1 (*Model statement and notation*)

Consider the multiple linear regression (MLR) model

$$y = \theta_0 + \theta_1 x_1 + \cdots + \theta_p x_p + \varepsilon, \quad \text{equivalently } \mathbf{y} = \mathbf{X}\boldsymbol{\theta} + \boldsymbol{\varepsilon}.$$

- (a) Define the dimensions of $\mathbf{y}$, $\mathbf{X}$, $\boldsymbol{\theta}$ and $\boldsymbol{\varepsilon}$ for a dataset with m observations and p predictors.
[4 points]
- (b) Explain why $\mathbf{X}$ typically contains a column of ones.
[2 points]
- (c) State the ordinary least squares (OLS) objective in both summation form and matrix norm form.
[4 points]

Question 2 (*Interpretation: “holding others fixed”*)

Suppose the fitted model for house price is

$$\widehat{\text{Price}} = \theta_0 + \theta_1 \cdot \text{Size} + \theta_2 \cdot \text{Age} + \theta_3 \cdot \text{Bedrooms}.$$

- (a) In words, interpret θ_1 and explicitly include the phrase “holding others fixed” (or equivalent).
[3 points]
- (b) Give one situation where interpreting θ_1 as a causal effect would be questionable.
[3 points]
- (c) If the model includes a predictor that is highly correlated with Size, what can happen to the sign and stability of $\hat{\theta}_1$? Explain briefly.
[4 points]

Question 3 (*Confounding example*)

You observe that ice-cream sales y increase with temperature x_1 . You later add a binary variable x_2 indicating weekend (1) vs weekday (0).

- (a) Explain qualitatively how omitting x_2 can bias the estimated slope of temperature.
[5 points]
- (b) Write the two models (with and without x_2) and describe what changes in the interpretation of the temperature coefficient when x_2 is included.
[5 points]

Question 4 (*Orthogonality vs. correlation*)

- (a) What does it mean for predictors to be orthogonal/uncorrelated in the context of regression design?
[3 points]
- (b) State one practical advantage of orthogonal predictors for *interpretation* and one for *numerical stability*.
[4 points]

- (c) Explain why high correlation can make $\mathbf{X}^\top \mathbf{X}$ nearly singular and what that implies for coefficient variance. [5 points]

Question 5 (“Who gets the credit?”)

Assume $x_2 \approx x_1$ in your dataset. Consider three models:

$$\hat{y} = 5 + 10x_1 + 0x_2, \quad \hat{y} = 5 + 0x_1 + 10x_2, \quad \hat{y} = 5 + 5x_1 + 5x_2.$$

- (a) Explain why these models can yield very similar predictions if $x_1 \approx x_2$. [4 points]
- (b) Which of the three models is easiest to *interpret* as a statement about the effect of x_1 alone? Justify. [3 points]
- (c) In one sentence, summarize the difference between “good prediction” and “good interpretation” in this setting. [3 points]

Question 6 (Short computation: OLS by normal equations)

You are given $m = 3$ observations with two predictors and an intercept:

$$\mathbf{X} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

(Here the predictors are columns 2 and 3.)

- (a) Compute $\mathbf{X}^\top \mathbf{X}$ and $\mathbf{X}^\top \mathbf{y}$. [6 points]
- (b) Solve the normal equations to obtain $\hat{\boldsymbol{\theta}}$. [6 points]
- (c) Interpret the coefficient associated with the *second* predictor (column 3). [3 points]

Question 7 (Perfect multicollinearity)

- (a) Give an example of a deterministic linear relationship among columns of $\mathbf{X}$ that creates *perfect* multicollinearity. [3 points]
- (b) What happens to the OLS solution when $\mathbf{X}^\top \mathbf{X}$ is singular? Mention both existence and uniqueness. [4 points]
- (c) Name one practical way to resolve perfect multicollinearity in a model. [3 points]

Question 8 (Categorical variables: dummy encoding)

A variable `Color` has three categories: `Red`, `Green`, `Blue`. You want to include it in a regression model with an intercept.

- (a) Construct a valid dummy/indicator encoding using $k - 1$ dummies. Specify the baseline. [5 points]
- (b) Write the regression model that includes the dummies and explain the meaning of the intercept under your encoding. [4 points]
- (c) Interpret the coefficient of the `IsRed` dummy. [3 points]

Question 9 (*The dummy variable trap*)

Suppose you instead include *all* k dummies for a categorical variable with k levels, and you include an intercept.

- (a) Prove (briefly) that the design matrix has linearly dependent columns. [5 points]
- (b) Explain what you would observe if you attempt to compute $(\mathbf{X}^\top \mathbf{X})^{-1}$. [3 points]
- (c) Give two fixes and explain which one you prefer in practice and why. [4 points]

Question 10 (*Interactions with a categorical variable*)

You model test score y as a function of study hours x and group membership D_B ($0 = \text{Group A}$, $1 = \text{Group B}$):

$$y = \theta_0 + \theta_1 x + \theta_2 D_B + \theta_3 (x \cdot D_B) + \varepsilon.$$

- (a) Write the fitted line (intercept and slope) for Group A and for Group B. [4 points]
- (b) Interpret θ_2 and θ_3 in words. [4 points]
- (c) In what real-world situation would adding the interaction term be justified? [3 points]

Question 11 (*Overfitting intuition*)

- (a) Explain, using a training vs. test error argument, why adding more predictors can *increase* test error. [5 points]
- (b) Give one symptom in the fitted coefficients that suggests overfitting or instability in OLS with many predictors. [3 points]

Question 12 (*Regularization: general idea*)

Consider the regularized objective

$$\min_{\boldsymbol{\theta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2 + \lambda \cdot \text{Penalty}(\boldsymbol{\theta}), \quad \lambda > 0.$$

- (a) What is the role of λ ? Describe what happens as $\lambda \rightarrow 0$ and as λ becomes very large. [4 points]
- (b) List three motivations for regularization (e.g., prediction, stability, interpretability) and connect each to the penalty term. [6 points]

Question 13 (*Ridge regression (L2)*)

Ridge regression solves

$$\min_{\boldsymbol{\theta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2 + \lambda \|\boldsymbol{\theta}\|_2^2.$$

- (a) Derive the ridge closed-form solution $\hat{\boldsymbol{\theta}}_{\text{ridge}} = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$. [7 points]
- (b) Why does adding $\lambda \mathbf{I}$ improve conditioning / invertibility? [4 points]
- (c) State (and briefly justify) whether ridge typically produces *exact* zeros in coefficients. [2 points]

Question 14 (*Lasso (L1)*)

Lasso solves

$$\min_{\boldsymbol{\theta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2 + \lambda \|\boldsymbol{\theta}\|_1.$$

- (a) Explain geometrically (in words) why L1 regularization can produce exact zeros. [5 points]
- (b) Give one advantage and one caveat of lasso when predictors are highly correlated. [4 points]
- (c) Why is there generally no simple closed-form solution for lasso? [3 points]

Question 15 (*L0 vs L1 vs L2: comparison table*)

Fill in a table comparing L0, L1 (lasso), and L2 (ridge) along the following axes:

- sparsity (does it set coefficients to zero?),
- computational difficulty,
- stability under correlated predictors,
- interpretability.

Then, in two sentences, explain when you would prefer each method. [10 points]

Question 16 (*Feature scaling: interpretation*)

- (a) Explain how changing the units of a predictor (e.g., meters to kilometers) changes the numerical value of its regression coefficient. [4 points]
- (b) Suppose θ_1 is the coefficient for **YearsExperience**. If you standardize **YearsExperience** to have mean 0 and standard deviation 1, how does the interpretation of the coefficient change? [4 points]

Question 17 (*Feature scaling: regularization*)

- (a) Explain why scaling is optional for OLS but important for ridge and lasso. [5 points]
- (b) Consider two predictors $x_1 \in [0, 1]$ and $x_2 \in [0, 1000]$. Without scaling, which coefficient tends to be penalized more by an L2 penalty, and why? [5 points]
- (c) State the critical rule about fitting a scaler when doing train/test evaluation. [2 points]

Question 18 (*Standardization vs min-max normalization*)

- (a) Write the formulas for z-score standardization and min-max normalization. [4 points]
- (b) Give one scenario where min-max normalization can be problematic, and explain why. [3 points]
- (c) If your goal is to compare effect sizes across predictors in a linear model, which scaling method is usually more convenient, and why? [3 points]

Question 19 (*Concept check: multiple choice*)

Circle **all** statements that are true.

- (a) If predictors are highly correlated, prediction can still be good even if individual coefficients are hard to interpret.
- (b) With an intercept in the model, a k -level categorical variable should be encoded with k dummies (one-hot) to avoid information loss.
- (c) Ridge regression can be viewed as OLS with an added constraint $\|\boldsymbol{\theta}\|_2^2 \leq C$ for some C .
- (d) Lasso always selects the same variable from a set of highly correlated predictors, regardless of small data changes.
- (e) Scaling features is essential before using lasso or ridge if you want the penalty to treat predictors fairly.

[5 points]

Question 20 (*Open-ended synthesis*)

You are given a dataset with: (i) 50 predictors, many of which are strongly correlated, (ii) only 200 observations, (iii) a need for an interpretable model that highlights a small set of important predictors.

- (a) Propose a modeling approach (OLS / ridge / lasso / elastic net / other) and justify your choice in terms of stability and sparsity. *[6 points]*
- (b) List two concrete preprocessing steps you would apply before fitting, and explain why. *[4 points]*